

Lec3: Epidemiological studies

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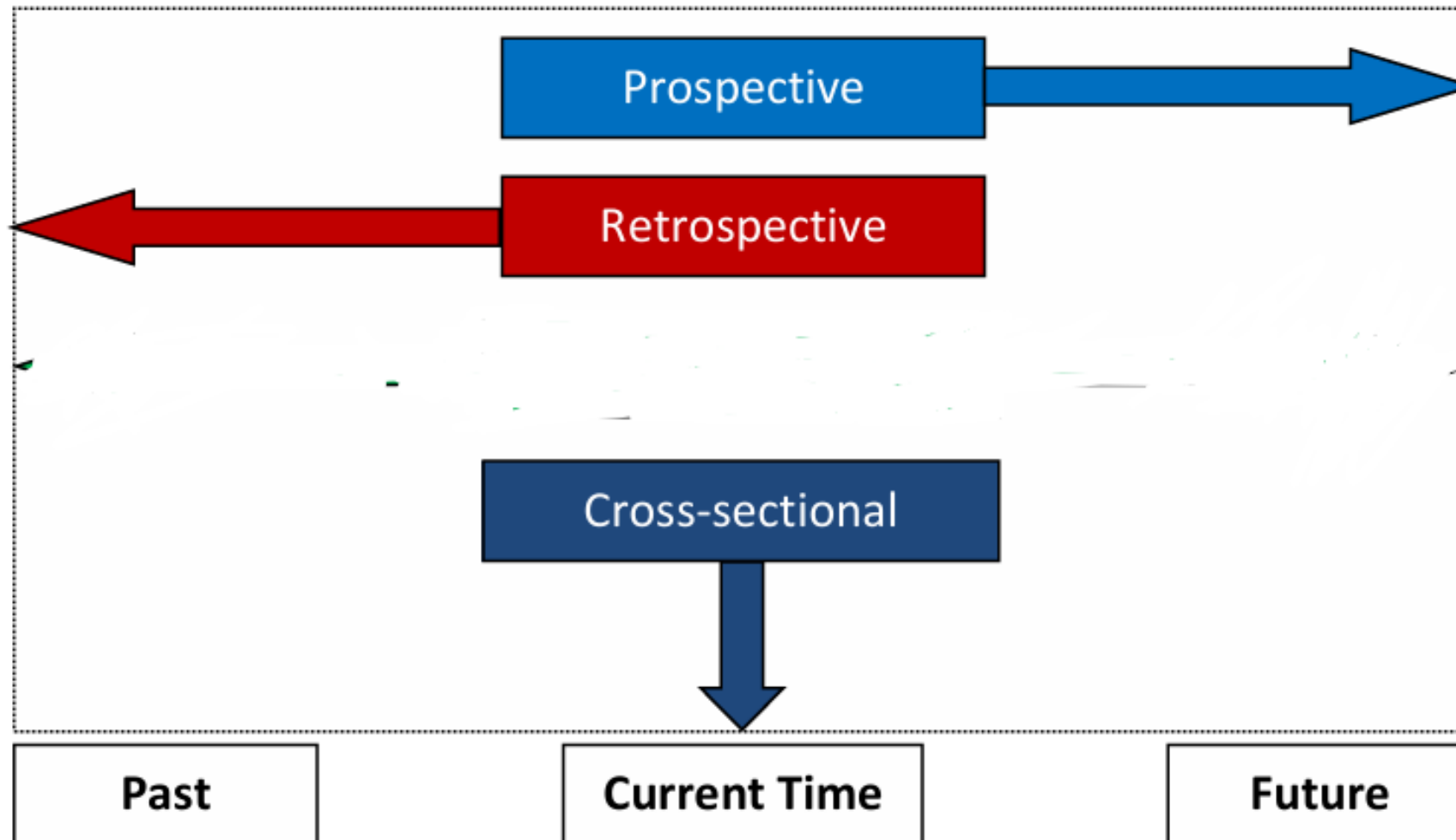
Epidemiological studies are organized into two divisions. These types of epidemiological studies complement one another, which are:

1- Observational epidemiology:

- 1- Descriptive study.
- 2- Analytical study.
 - a- Case-control study.
 - b- Cohort study.

2- Experimental epidemiology (clinical trials):

- 1- Randomized control trials.
- 2- Field trials.
- 3- Community trials.



1- Observational epidemiology.

1- Descriptive study:

It is one that attempts to do no more than **describe the pattern of occurrence** of a condition relative to other characteristics of the population. For example, a study that attempted to measure the degree of dental caries in a school relative to the age, gender, and socioeconomic characteristics of the children without attempting to explain why those particular distributions were found. Descriptive studies are usually the first phase of any epidemiological investigation.

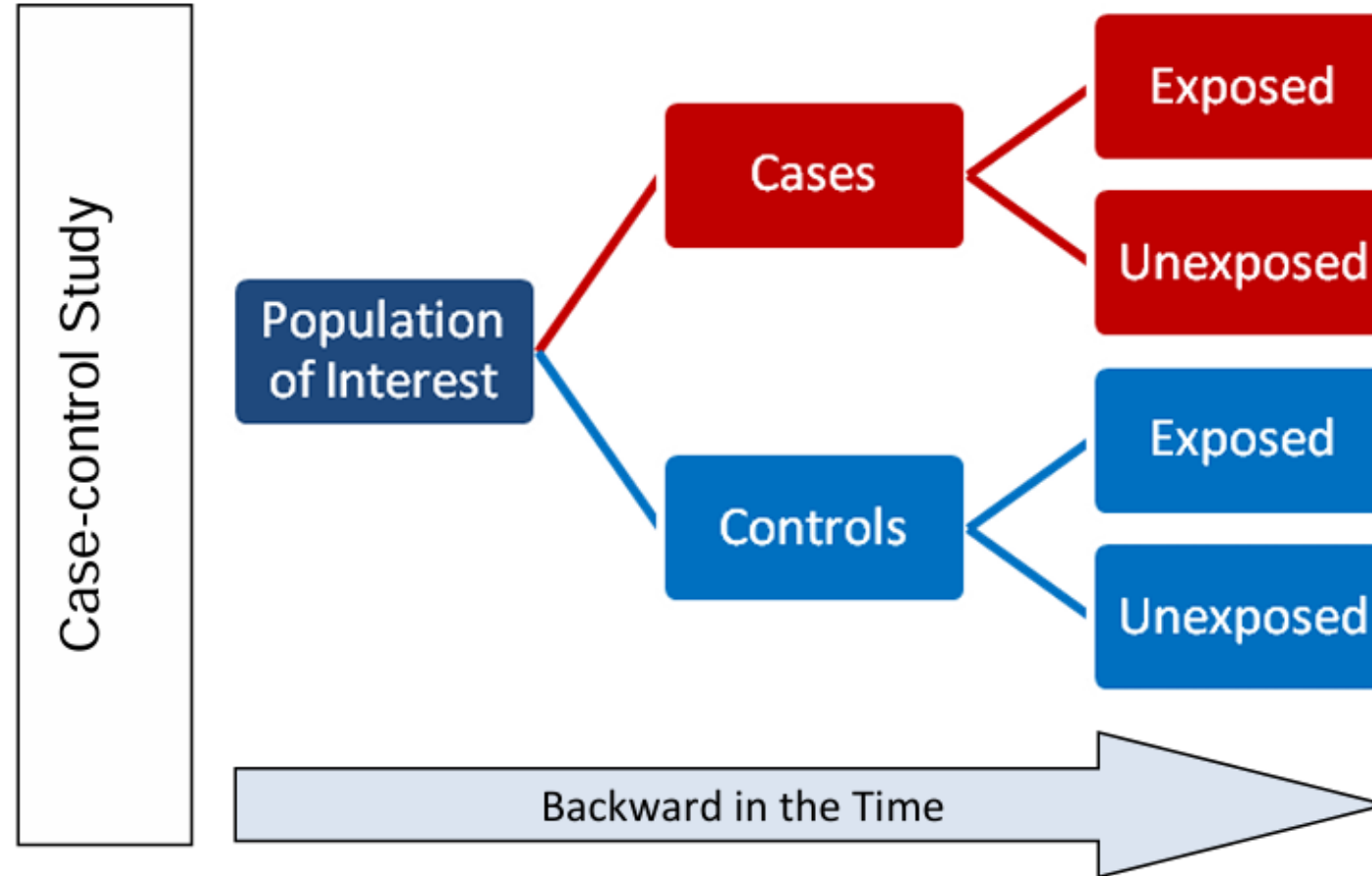
2- Analytical study:

It is one that attempts to explain the distribution of a condition relative to some other factor. **It is most often used in studies to determine the etiology of the disease.** For example, measuring the influence of water fluoridation on the distribution of dental caries in a population.

a- Case-control study.

b- Cohort study.

Types of analytical studies:



Types of analytical studies:

a- Case-control study: it is often called retrospective study used to test hypothesis. The case-control studies have the following distinct features:

1. Both exposure and outcome of the disease have **occurred before** the start of the study.
2. The study proceeds **backward** from effect to cause.
3. It uses a control group for comparison.

There are four basic steps in conducting a case control study:

1. Selection of cases and controls.

Definition of a case:

The diagnostic criteria of the disease and stage, if any, to be included in the study must be specified before the study is undertaken and should not be altered or changed until the study is over.

Selection of controls:

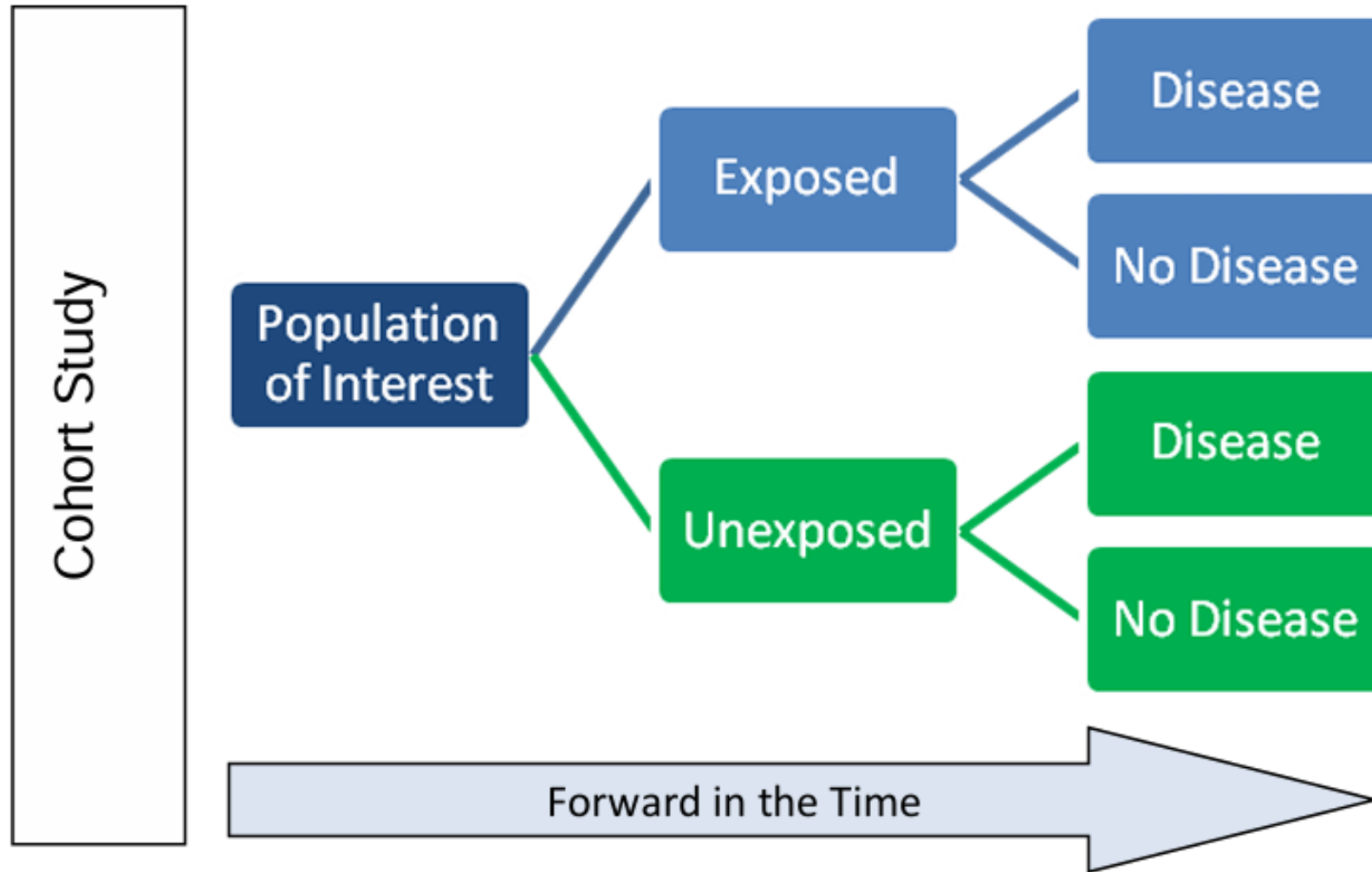
Controls must be free from the disease under study. Similar to the cases, except for the absence of disease.

2.matching:is the process by which one selects controls in way that they are similar to cases with regard to certain pertinent selected variables(eg.age) which are known to influence the outcome of the disease and if not matched could distort the results.

3. Measurement of exposure: this may be obtained by interviews, questionnaire or studying past records.

4. Analysis and interpretation.

Types of analytical studies:



Types of analytical studies:

B- Cohort study is known by a **number of names**: Prospective study, longitudinal study, incidence study, forward-looking study.

this type of analytical study which is usually undertaken to refute or support the existence of an association between suspected causes and the disease, it has the following features:

1. The cohorts are identified **prior** to the appearance of the disease under investigation.
2. Study groups are **followed** through time to see which members develop the disease.
3. The study **proceeds forwards** from cause to effect (prospective study).

Elements of cohort study:

1. Selection of study subjects.
2. Obtaining data on exposure.
3. Selection of comparison groups.
4. Follow up
5. Analysis.

The differences between Case-control and Cohort studies

Case-control study

- 1- Retrospective
- 2- Disease has already occurred
- 3- Presence of exposure in cases and controls compared
- 4- Relatively easy to carry out
- 5- Useful for rare cases
- 6- Relatively less costly
- 7- No dropouts

Cohort study

- 1- Prospective
- 2- Disease is expected to occur in future
- 3- Development of disease in exposed and non exposed compared
- 4- Time consuming and difficult to carry out
- 5- Suitable for common disease
- 6- Expensive
- 7- Dropout rate higher

2- Experimental epidemiology.

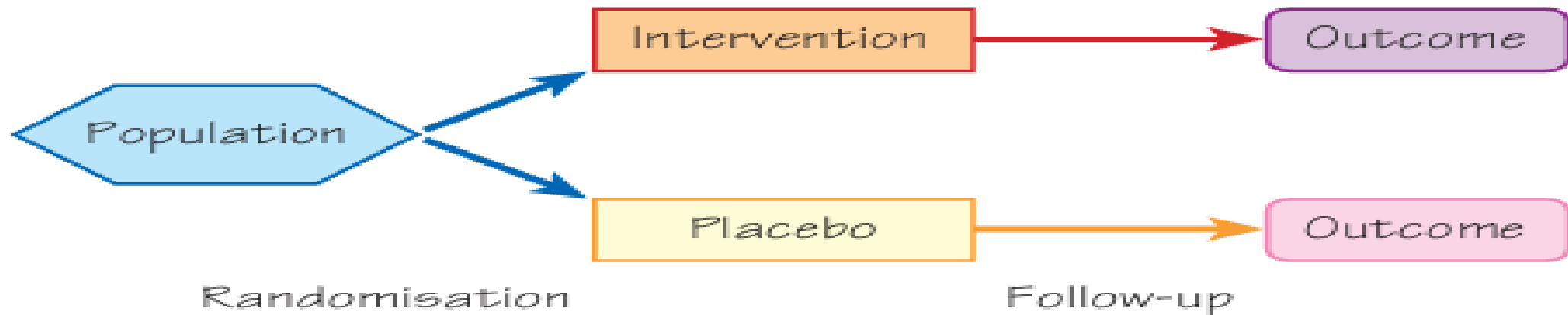
Experimental studies are ones where researchers introduce an intervention and study the effects. Experimental studies are usually randomized, meaning the subjects are grouped by chance. Clinical trials are designed to bring out the most appropriate intervention for patients suffering from a given condition. When a new treatment has been developed, it is important to provide the answers to two questions:

1- Does it work?

2- Is it better than the existing treatment?

The studies under this category are as follows:

1- Randomized control trails: When the investigation is to be carried out in a group of individuals, a parallel group not exposed must also be studied in the same way; this called the control group and should be matched the group under study. It is important to supply the control group with a substance similar in appearance and other properties to that being test; **this is called a placebo**. It is always desirable that the investigator should not know whether a subject is a member of test or a control group(**the study should be blind**). If the subject is also in ignorance of whether he is using a test product or placebo, the study is termed **double blind**. This is to avoid unconscious bias in the diagnosis.



2- Field trials (randomized uncontrolled):

a- Preventive trials: It is used to estimate the effect of preventive techniques like vaccination.

b- Risk factor trials: It is used when specific risk factors are averted in groups of population, like dietary pattern. Both randomized control trials and field trials involve individual units of population.

3- Community trials (non-randomized controlled): Such studies are the only way in which a general intervention like fluoridation has to be tried out for reducing dental caries. Control communities in the neighborhood can be selected for comparison.

Ethical issues The practice of epidemiology requires basic principles of ethics, these are:

1. People who have been exposed to a health hazard should know that the studies carried out on them may not improve their personal situation but may help to protect thousands of other people.
2. Informed consent must be obtained from participants in studies and they must retain the right to **withdraw at any time**.
3. Epidemiologists must respect personal privacy and confidentiality at all times.
4. All proposals for epidemiological studies should be submitted to institutional ethics committees before work begins.